Chapter 5. Entire Functions

Jensen's formula, order of growth, infinite products, Weierstrass and Hadamard factorization

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Overview

Entire functions are holomorphic in the whole complex plane. The chapter is organized around three questions.

1. *Where can they vanish?* The obvious necessary condition is sufficient: for any sequence $\{z_n\}$ with no limit point in $\mathbb{C}$ there is an entire function vanishing exactly there. The construction is inspired by Euler's product for $\sin \pi z$, refined by the *Weierstrass canonical factors*.
2. *How do they grow at infinity?* The larger a function is, the more zeros it can have; for a polynomial of degree d, $\sup_{|z|=R}|P(z)| \approx R^d$ and P has d zeros. A precise version is *Jensen's formula*, linking the number of zeros in a disc to the logarithmic average of $|f|$ on the circle.
3. *To what extent are they determined by their zeros?* If f has finite order, it is determined by its zeros up to a simple factor: the *Hadamard factorization theorem*.

1 Jensen's formula

Motivation. How many zeros can an entire function have in a disc of radius R? Jensen's formula relates the zeros of f to the average of $\log|f|$ on the boundary circle, so growth controls the number of zeros.

D_R and C_R denote the open disc and circle of radius R centered at 0. We exclude the function that vanishes identically.

Theorem 1.1 (Jensen's formula)

Let Ω be an open set that contains the closure of a disc D_R and suppose that f is holomorphic in Ω, $f(0) \neq 0$, and f vanishes nowhere on the circle C_R. If $z_1, \dots, z_N$ denote the zeros of f inside the disc (counted with multiplicities), then

$$\log|f(0)| = \sum_{k=1}^{N} \log\left(\frac{|z_k|}{R}\right) + \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\, d\theta. \tag{1}$$

Proof.

Step 1. If f_1, f_2 satisfy the hypotheses and (1), so does $f_1 f_2$, since $\log xy = \log x + \log y$ for $x, y > 0$ and the zeros of $f_1 f_2$ are the union of those of f_1 and f_2.

Step 2. $g(z) = f(z)/((z - z_1)\cdots(z - z_N))$, defined on $\Omega - \{z_1, \dots, z_N\}$, is bounded near each z_j, so each z_j is a removable singularity: $f(z) = (z - z_1)\cdots(z - z_N)g(z)$ with g holomorphic in Ω and nowhere vanishing on $\overline{D_R}$. By Step 1 it suffices to prove (1) for such g and for functions $z - z_j$.

Step 3. For g nowhere vanishing on $\overline{D_R}$ we must show $\log|g(0)| = \frac{1}{2\pi}\int_0^{2\pi} \log|g(Re^{i\theta})|\, d\theta$. In a slightly larger disc, $g = e^h$ with h holomorphic (discs are simply connected; Theorem 6.2). Then $|g| = e^{\mathrm{Re}(h)}$, so $\log|g| = \mathrm{Re}(h)$, and the mean value property (Corollary 7.3) gives the formula.

Step 4. For $f(z) = z - w$, $w \in D_R$, we must show $\log|w| = \log\frac{|w|}{R} + \frac{1}{2\pi}\int_0^{2\pi} \log|Re^{i\theta} - w|\, d\theta$. Since $\log(|w|/R) = \log|w| - \log R$ and $\log|Re^{i\theta} - w| = \log R + \log|e^{i\theta} - w/R|$, it suffices to show $\int_0^{2\pi} \log|e^{i\theta} - a|\, d\theta = 0$ for $|a| < 1$, equivalently (via $\theta \mapsto -\theta$, $|e^{i\theta} - a| = |1 - ae^{-i\theta}|$) $\int_0^{2\pi} \log|1 - ae^{i\theta}|\, d\theta = 0$. $F(z) = 1 - az$ vanishes nowhere on the closed unit disc, so $F = e^G$ in a disc of radius > 1 and $\log|F| = \mathrm{Re}(G)$ is harmonic. Since $\log|F(0)| = 0$, the mean value property gives the result. ∎

For f holomorphic on $\overline{D_R}$, let $n(r) = n_f(r)$ be the number of zeros of f (with multiplicities) in D_r,

$0 < r < R$; $n(r)$ is non-decreasing. If $f(0) \neq 0$ and f does not vanish on C_R, then

$$\int_0^R n(r)\frac{dr}{r} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(Re^{i\theta})|\,d\theta - \log|f(0)|, \tag{2}$$

which follows from Jensen's formula and:

Lemma 1.2

If $z_1, \ldots, z_N$ are the zeros of f inside the disc D_R, then

$$\int_0^R n(r)\frac{dr}{r} = \sum_{k=1}^{N} \log\left|\frac{R}{z_k}\right|.$$

Proof. $\sum_{k=1}^N \log\left|\frac{R}{z_k}\right| = \sum_{k=1}^N \int_{|z_k|}^R \frac{dr}{r}$. With $\eta_k(r) = 1$ if $r > |z_k|$ and 0 otherwise, $\sum_k \eta_k(r) = n(r)$, so $\sum_k \int_{|z_k|}^R \frac{dr}{r} = \sum_k \int_0^R \eta_k(r)\frac{dr}{r} = \int_0^R n(r)\frac{dr}{r}$. ∎

2 Functions of finite order

Motivation. Measuring growth by an order ρ with $|f(z)| \leq Ae^{B|z|^\rho}$, Jensen's formula gives a quantitative bound on the number of zeros. This prepares the factorization theorems of the chapter.

Definition (order of growth)

An entire function f has an **order of growth** $\leq \rho$ if $|f(z)| \leq Ae^{B|z|^\rho}$ for all $z \in \mathbb{C}$, for some constants $A, B > 0$. The **order of growth** of f is $\rho_f = \inf \rho$ over all such $\rho > 0$.

For example, e^{z^2} has order of growth 2.

Theorem 2.1

If f is an entire function that has an order of growth $\leq \rho$, then:
(i) $n(r) \leq Cr^\rho$ for some $C > 0$ and all sufficiently large r.
(ii) If $z_1, z_2, \ldots$ denote the zeros of f, with $z_k \neq 0$, then for all $s > \rho$ we have $\displaystyle\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty$.

Proof. It suffices to treat $f(0) \neq 0$: otherwise use $F(z) = f(z)/z^\ell$, ℓ the order of the zero at 0; n_f and n_F differ by a constant and F also has order of growth $\leq \rho$. By (2) with $R = 2r$,

$$\int_r^{2r} n(x)\frac{dx}{x} \leq \int_0^{2r} n(x)\frac{dx}{x} = \frac{1}{2\pi}\int_0^{2\pi} \log|f(2re^{i\theta})|\,d\theta - \log|f(0)|.$$

Since n is increasing, the left side is $\geq n(r)\int_r^{2r}\frac{dx}{x} = n(r)\log 2$; and by the growth condition $\int_0^{2\pi}\log|f(2re^{i\theta})|\,d\theta \leq \int_0^{2\pi}\log|Ae^{B(2r)^\rho}|\,d\theta \leq C'r^\rho$ for large r. Hence $n(r) \leq Cr^\rho$. For (ii),

$$\sum_{|z_k|\geq 1} |z_k|^{-s} = \sum_{j=0}^{\infty} \sum_{2^j \leq |z_k| < 2^{j+1}} |z_k|^{-s} \leq \sum_{j=0}^{\infty} 2^{-js} n(2^{j+1}) \leq c\sum_{j=0}^{\infty} 2^{-js}2^{(j+1)\rho} \leq c'\sum_{j=0}^{\infty}(2^{\rho-s})^j < \infty,$$

since $s > \rho$. ∎

The condition $s > \rho$ cannot be improved:

Example 1

$f(z) = \sin \pi z = \frac{e^{i\pi z} - e^{-i\pi z}}{2i}$ satisfies $|f(z)| \le e^{\pi|z|}$, and taking $z = ix$ shows the order of growth is exactly 1. f vanishes to order 1 at each $n \in \mathbb{Z}$, and $\sum_{n\neq 0} 1/|n|^s < \infty$ precisely when $s > 1$.

Example 2

$f(z) = \cos z^{1/2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{(2n)!}$ is entire with $|f(z)| \le e^{|z|^{1/2}}$, of order $1/2$. It vanishes at $z_n = ((n + 1/2)\pi)^2$, and $\sum_n 1/|z_n|^s < \infty$ exactly when $s > 1/2$.

Given $z_1, z_2, \ldots$, is there an entire f with zeros exactly there? Necessarily $\lim_{k\to\infty} |z_k| = \infty$ (Theorem 4.8). Weierstrass showed this suffices, by inserting factors into $(z-z_1)(z-z_2)\cdots$ that guarantee convergence without introducing new zeros.

3 Infinite products

Motivation. A polynomial is determined by its zeros up to a constant, through $\prod(z - a_k)$. To do the same for entire functions with infinitely many zeros we need infinite products and conditions for their convergence.

3.1 Generalities

For complex a_n, the product $\prod_{n=1}^{\infty}(1 + a_n)$ **converges** if $\lim_{N\to\infty} \prod_{n=1}^{N}(1 + a_n)$ exists.

Proposition 3.1

If $\sum |a_n| < \infty$, then the product $\prod_{n=1}^{\infty}(1 + a_n)$ converges. Moreover, the product converges to 0 if and only if one of its factors is 0.

Proof. For large n, $|a_n| < 1/2$; discarding finitely many terms, assume this for all n. Define $\log(1 + a_n)$ by the power series (6) of Chapter 3, so $1 + z = e^{\log(1+z)}$ for $|z| < 1$. Then $\prod_{n=1}^{N}(1 + a_n) = \prod_{n=1}^{N} e^{\log(1+a_n)} = e^{B_N}$ with $B_N = \sum_{n=1}^{N} b_n$, $b_n = \log(1 + a_n)$. From the power series, $|\log(1 + z)| \le 2|z|$ for $|z| < 1/2$, so $|b_n| \le 2|a_n|$ and $B_N \to B$. By continuity of the exponential, $e^{B_N} \to e^B$. If $1 + a_n \neq 0$ for all n, the limit is $e^B \neq 0$. ∎

Proposition 3.2

Suppose $\{F_n\}$ is a sequence of holomorphic functions on the open set Ω. If there exist constants $c_n > 0$ such that $\sum c_n < \infty$ and $|F_n(z) - 1| \le c_n$ for all $z \in \Omega$, then:
(i) The product $\prod_{n=1}^{\infty} F_n(z)$ converges uniformly in Ω to a holomorphic function $F(z)$.
(ii) If $F_n(z)$ does not vanish for any n, then

$$\frac{F'(z)}{F(z)} = \sum_{n=1}^{\infty} \frac{F_n'(z)}{F_n(z)}.$$

Proof. (i) Write $F_n(z) = 1 + a_n(z)$ with $|a_n(z)| \le c_n$ and argue as in Proposition 3.1; the estimates are uniform in z since the c_n are constants. The uniform limit is holomorphic (Theorem 5.2).

(ii) Let $K \subset \Omega$ be compact and $G_N = \prod_{n=1}^{N} F_n$. Since $G_N \to F$ uniformly, Theorem 5.3 gives $G_N' \to F'$ uniformly on K. G_N is uniformly bounded below on K, so $G_N'/G_N \to F'/F$ uniformly on K; and $G_N'/G_N = \sum_{n=1}^{N} F_n'/F_n$ (Section 4). ∎

3.2 Example: the product formula for the sine function

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \Big(1 - \frac{z^2}{n^2}\Big) \tag{3}$$

for all $z \in \mathbb{C}$; it is derived from the sum formula for the cotangent

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_{N\to\infty} \sum_{|n|\le N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2}, \tag{4}$$

valid for $z \notin \mathbb{Z}$. The halves of $\sum 1/(z+n)$ with $n > 0$ and $n < 0$ diverge separately; only the symmetric interpretation converges.

Proof of (4). $F(z) = \pi \cot \pi z$ has the properties:

(i) $F(z+1) = F(z)$ for $z \notin \mathbb{Z}$;

(ii) $F(z) = \frac{1}{z} + F_0(z)$ with F_0 analytic near 0;

(iii) F has simple poles at the integers and no other singularities. The function $\sum_{n=-\infty}^{\infty} \frac{1}{z+n} = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ has the same three properties: for (i),

$$\sum_{|n|\le N} \frac{1}{z+1+n} = \frac{1}{z+1+N} - \frac{1}{z-N} + \sum_{|n|\le N} \frac{1}{z+n},$$

and let $N \to \infty$; (ii), (iii) are evident from $\frac{1}{z} + \sum_{n\ge 1} \frac{2z}{z^2-n^2}$. Hence $\Delta(z) = F(z) - \sum \frac{1}{z+n}$ is periodic with period 1, has a removable singularity at 0 by (ii), hence at all integers by periodicity, and is entire. It suffices to show Δ is bounded, hence (by periodicity) bounded in the strip $|\mathrm{Re}(z)| \le 1/2$. It is bounded in the rectangle $|\mathrm{Im}(z)| \le 1$. For $z = x + iy$ with $y > 1$,

$$\pi \cot \pi z = \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \pi i \frac{e^{-2\pi y} + e^{-2\pi i x}}{e^{-2\pi y} - e^{-2\pi i x}}$$

is bounded, and

$$\Big|\frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}\Big| = \Big|\frac{1}{x+iy} + \sum_{n=1}^{\infty} \frac{2(x+iy)}{x^2 - y^2 - n^2 + 2ixy}\Big| \le C + C\sum_{n=1}^{\infty} \frac{y}{y^2+n^2}.$$

The last sum is majorized by $\int_0^{\infty} \frac{y}{y^2+x^2}\,dx$ (the integrand is decreasing in x), which is independent of y (change variables $x \mapsto yx$). Similarly for $y < -1$. So Δ is bounded on $\mathbb{C}$, hence constant by Liouville; Δ is odd, so the constant is 0. ∎

Proof of (3). Let $G(z) = \sin \pi z$ and $P(z) = \pi z \prod_{n=1}^{\infty}(1 - z^2/n^2)$. Since $\sum 1/n^2 < \infty$, Proposition 3.2 shows P converges and, away from the integers, $\frac{P'(z)}{P(z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2-n^2}$. Since $G'/G = \pi \cot \pi z$, (4) gives

$$\Big(\frac{P}{G}\Big)' = \frac{P}{G}\Big[\frac{P'}{P} - \frac{G'}{G}\Big] = 0,$$

so $P = cG$. Dividing by z and letting $z \to 0$, $c = 1$. ∎

4 Weierstrass infinite products

Motivation. The naive product $\prod(1 - z/a_n)$ may diverge. Weierstrass's canonical factors repair this, so that an entire function with any prescribed sequence of zeros (tending to infinity) exists.

Theorem 4.1 (Weierstrass)

Given any sequence $\{a_n\}$ of complex numbers with $|a_n| \to \infty$ as $n \to \infty$, there exists an entire function f that vanishes at all $z = a_n$ and nowhere else. Any other such entire function is of the form $f(z)e^{g(z)}$, where g is entire.

Repetitions are allowed in $\{a_n\}$, so this gives entire functions with prescribed zeros *and multiplicities* (the multiplicity of a zero a is the m with $f(z) = (z-a)^m g(z)$, $g(a) \neq 0$).

Proof. If f_1, f_2 both vanish exactly at the a_n (with multiplicities), f_1/f_2 has removable singularities there, so it is entire and nowhere vanishing; hence $f_1/f_2 = e^g$ for an entire g (Theorem 6.2). This proves the last statement.

To construct f: the naive product $\prod(1 - z/a_n)$ converges only for suitable $\{a_n\}$, so we insert exponential factors. Define the **canonical factors** of degree k:

$$E_0(z) = 1 - z, \qquad E_k(z) = (1-z)e^{z+z^2/2+\cdots+z^k/k} \quad (k \geq 1).$$

Suppose f is to have a zero of order m at the origin and $a_1, a_2, \ldots$ are non-zero. The **Weierstrass product** is

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Fix $R > 0$ and $|z| < R$. Factors with $|a_n| \leq 2R$ are finitely many, and their product vanishes exactly at the a_n with $|a_n| < R$. If $|a_n| > 2R$ then $|z/a_n| \leq 1/2$, and by Lemma 4.2 (with c independent of n)

$$|1 - E_n(z/a_n)| \leq c\Big|\frac{z}{a_n}\Big|^{n+1} \leq \frac{c}{2^{n+1}}.$$

By Section 3, $\prod_{|a_n|>2R} E_n(z/a_n)$ is holomorphic and non-vanishing in $|z| < R$. As R is arbitrary, f has the required properties. ∎

Lemma 4.2

If $|z| \leq 1/2$, then $|1 - E_k(z)| \leq c|z|^{k+1}$ for some $c > 0$.

Proof. For $|z| \leq 1/2$, $1 - z = e^{\log(1-z)}$ (power series), so $E_k(z) = e^{\log(1-z)+z+z^2/2+\cdots+z^k/k} = e^w$ with $w = -\sum_{n=k+1}^{\infty} z^n/n$. Since $|z| \leq 1/2$,

$$|w| \leq |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \leq |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \leq 2|z|^{k+1}.$$

In particular $|w| \leq 1$, so $|1 - E_k(z)| = |1 - e^w| \leq c'|w| \leq c|z|^{k+1}$. ∎

Remark

The constant c is independent of k: one may take $c' = e$, $c = 2e$.

5 Hadamard's factorization theorem

Weierstrass's theorem writes a function with zeros $a_1, a_2, \ldots$ as $e^{g(z)} z^m \prod E_n(z/a_n)$. Hadamard showed that for functions of finite order the degree of the canonical factors can be taken constant, and g is then a polynomial. Recall that if f has order of growth $\leq \rho$, then $n(r) \leq Cr^\rho$ for large r and $\sum |a_n|^{-s} < \infty$ for $s > \rho$ (Theorem 2.1).

Theorem 5.1 (Hadamard)

Suppose f is entire and has growth order ρ_0. Let k be the integer so that $k \le \rho_0 < k+1$. If $a_1, a_2, \dots$ denote the (non-zero) zeros of f, then

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\le k$, and m is the order of the zero of f at $z = 0$.

Main lemmas.

Lemma 5.2

The canonical products satisfy

$$|E_k(z)| \ge e^{-c|z|^{k+1}} \text{ if } |z| \le 1/2 \qquad \text{and} \qquad |E_k(z)| \ge |1-z| e^{-c'|z|^k} \text{ if } |z| \ge 1/2.$$

Proof. If $|z| \le 1/2$, $E_k(z) = e^{\log(1-z) + \sum_{n=1}^{k} z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w$, and $|e^w| \ge e^{-|w|}$ with $|w| \le c|z|^{k+1}$. If $|z| \ge 1/2$, $|E_k(z)| = |1-z||e^{z+z^2/2+\cdots+z^k/k}|$, and $|e^{z+\cdots+z^k/k}| \ge e^{-|z+\cdots+z^k/k|} \ge e^{-c'|z|^k}$ (as $|z| \ge 1/2$). ■

The key is a lower bound for the product of canonical factors away from the zeros.

Lemma 5.3

For any s with $\rho_0 < s < k+1$, we have

$$\Big| \prod_{n=1}^{\infty} E_k(z/a_n) \Big| \ge e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \dots$.

Proof. Write $\prod_n E_k(z/a_n) = \prod_{|a_n| \le 2|z|} E_k(z/a_n) \prod_{|a_n| > 2|z|} E_k(z/a_n)$. For the second product, with no restriction on z, Lemma 5.2 gives

$$\Big| \prod_{|a_n|>2|z|} E_k(z/a_n) \Big| \ge \prod_{|a_n|>2|z|} e^{-c|z/a_n|^{k+1}} = e^{-c|z|^{k+1} \sum_{|a_n|>2|z|} |a_n|^{-k-1}},$$

and since $|a_n| > 2|z|$ and $s < k+1$, $|a_n|^{-k-1} = |a_n|^{-s}|a_n|^{s-k-1} \le C|a_n|^{-s}|z|^{s-k-1}$; as $\sum |a_n|^{-s} < \infty$, this product is $\ge e^{-c|z|^s}$. For the first product, by the second part of Lemma 5.2,

$$\Big| \prod_{|a_n| \le 2|z|} E_k(z/a_n) \Big| \ge \prod_{|a_n| \le 2|z|} \Big|1 - \frac{z}{a_n}\Big| \prod_{|a_n| \le 2|z|} e^{-c'|z/a_n|^k}. \tag{5}$$

Here $\prod_{|a_n| \le 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k \sum_{|a_n| \le 2|z|} |a_n|^{-k}}$ and $|a_n|^{-k} = |a_n|^{-s}|a_n|^{s-k} \le C|a_n|^{-s}|z|^{s-k}$, so this is $\ge e^{-c|z|^s}$. For the first factor of (5) we need the restriction on z: if z is not in the disc of radius $|a_n|^{-k-1}$ about a_n, then $|a_n - z| \ge |a_n|^{-k-1}$, so

$$\prod_{|a_n| \le 2|z|} \Big|1 - \frac{z}{a_n}\Big| = \prod_{|a_n| \le 2|z|} \Big|\frac{a_n - z}{a_n}\Big| \ge \prod_{|a_n| \le 2|z|} |a_n|^{-k-1}|a_n|^{-1} = \prod_{|a_n| \le 2|z|} |a_n|^{-k-2}.$$

Finally $\log \prod_{|a_n| \le 2|z|} |a_n|^{k+2} = (k+2) \sum_{|a_n| \le 2|z|} \log|a_n| \le (k+2) n(2|z|) \log 2|z| \le c|z|^s \log 2|z| \le c'|z|^{s'}$ for any $s' > s$, using $n(2|z|) \le c|z|^s$ (Theorem 2.1). Since $s > \rho_0$ was arbitrary, starting with s close enough to ρ_0 gives the lemma (with s' in place of s). ■

Corollary 5.4

There exists a sequence of radii $r_1, r_2, \dots$ with $r_m \to \infty$, such that

$$\Big|\prod_{n=1}^{\infty} E_k(z/a_n)\Big| \geq e^{-c|z|^s} \qquad \text{for } |z| = r_m.$$

Proof. Since $\sum |a_n|^{-k-1} < \infty$, there is N with $\sum_{n\geq N} |a_n|^{-k-1} < 1/10$. Given consecutive large integers L, $L+1$, there is $r \in [L, L+1]$ such that the circle of radius r about 0 meets none of the forbidden discs of Lemma 5.3: otherwise the intervals $I_n = [|a_n| - |a_n|^{-k-1}, |a_n| + |a_n|^{-k-1}]$, of length $2|a_n|^{-k-1}$, would cover $[L, L+1]$, giving $2\sum_{n\geq N} |a_n|^{-k-1} \geq 1$, a contradiction. Apply Lemma 5.3 with $|z| = r$. ∎

Proof of Hadamard's theorem. Let $E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n)$. E is entire, as in the proof of Theorem 4.1: by Lemma 4.2, $|1 - E_k(z/a_n)| \leq c|z/a_n|^{k+1}$ for large n, and $\sum |a_n|^{-k-1}$ converges ($\rho_0 < s < k+1$). E has the zeros of f, so f/E is entire and nowhere vanishing: $f(z)/E(z) = e^{g(z)}$ for an entire g. Since f has growth order ρ_0 and by Corollary 5.4,

$$e^{\operatorname{Re}(g(z))} = \Big|\frac{f(z)}{E(z)}\Big| \leq c' e^{c|z|^s} \qquad \text{for } |z| = r_m,$$

so $\operatorname{Re}(g(z)) \leq C|z|^s$ for $|z| = r_m$. Lemma 5.5 gives that g is a polynomial of degree $\leq s$, hence $\leq k$ (as $s < k+1$). ∎

Lemma 5.5

Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies $u(z) \leq Cr^s$ whenever $|z| = r$, for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\leq s$.

Proof. Write $g(z) = \sum_{n=0}^{\infty} a_n z^n$. By Theorem 7.1,

$$\frac{1}{2\pi}\int_0^{2\pi} g(re^{i\theta}) e^{-in\theta}\, d\theta = \begin{cases} a_n r^n & n \geq 0, \\ 0 & n < 0. \end{cases} \tag{6}$$

Taking complex conjugates, $\frac{1}{2\pi}\int_0^{2\pi} \overline{g(re^{i\theta})} e^{-in\theta}\, d\theta = 0$ for $n > 0$ (7). Since $2u = g + \overline{g}$, adding (6) and (7),

$$a_n r^n = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta}) e^{-in\theta}\, d\theta \quad (n > 0), \qquad 2\operatorname{Re}(a_0) = \frac{1}{\pi}\int_0^{2\pi} u(re^{i\theta})\, d\theta.$$

[Supplement]Since $\int_0^{2\pi} e^{-in\theta} d\theta = 0$ for $n > 0$, $a_n r^n = -\frac{1}{\pi}\int_0^{2\pi} [Cr^s - u(re^{i\theta})] e^{-in\theta}\, d\theta$, and $Cr^s - u \geq 0$ on the circles in question, so

$$|a_n| r^n \leq \frac{1}{\pi}\int_0^{2\pi} [Cr^s - u(re^{i\theta})]\, d\theta = 2Cr^s - 2\operatorname{Re}(a_0).$$

Letting $r \to \infty$ along the sequence, $a_n = 0$ for $n > s$. ∎

6 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 5 (book numbers kept): Wallis's product from the sine product, the order of the Jacobi theta function, and two applications of Hadamard's theorem.

Exercise 5.6 (Wallis's product)

(a) Prove Wallis's product formula

$$\frac{\pi}{2} = \prod_{m=1}^{\infty} \frac{2m \cdot 2m}{(2m-1)(2m+1)} = \frac{2 \cdot 2}{1 \cdot 3} \cdot \frac{4 \cdot 4}{3 \cdot 5} \cdots .$$

(b) Deduce that $\lim_{n\to\infty} \dfrac{2^{2n}(n!)^2}{(2n)!\,(2n+1)^{1/2}} = \sqrt{\pi/2}$.

Solution.

(a) Put $z = 1/2$ in the product formula (3) for the sine:

$$1 = \sin\frac{\pi}{2} = \frac{\pi}{2} \prod_{m=1}^{\infty} \Big(1 - \frac{1}{4m^2}\Big).$$

So the partial products $P_N = \prod_{m\le N}(1 - \frac{1}{4m^2})$ converge to $2/\pi \neq 0$. Hence $1/P_N \to \pi/2$, and

$$\frac{1}{P_N} = \prod_{m=1}^{N} \frac{4m^2}{4m^2-1} = \prod_{m=1}^{N} \frac{2m \cdot 2m}{(2m-1)(2m+1)}.$$

(b) Let $W_N = 1/P_N$.

Step 1. We have $\prod_{m\le N} 2m = 2^N N!$. The odd numbers below $2N$ and below $2N+2$ have products

$$\prod_{m\le N} (2m-1) = \frac{(2N)!}{2^N N!}, \qquad \prod_{m\le N} (2m+1) = \frac{(2N+1)!}{2^N N!}.$$

Step 2. Therefore

$$W_N = \frac{(2^N N!)^4}{(2N)!\,(2N+1)!} = \frac{2^{4N}(N!)^4}{((2N)!)^2(2N+1)} = \Big(\frac{2^{2N}(N!)^2}{(2N)!\,(2N+1)^{1/2}}\Big)^2.$$

Step 3. The bracket is positive and its square tends to $\pi/2$ by (a), so it tends to $\sqrt{\pi/2}$. ∎

Exercise 5.3 (Order of the theta function)

Fix τ with $t = \operatorname{Im}\tau > 0$. Show that the Jacobi theta function

$$\Theta(z \mid \tau) = \sum_{n=-\infty}^{\infty} e^{i\pi n^2 \tau} e^{2\pi i n z}$$

is an entire function of z of order exactly 2.

Solution.

Step 1. (Upper bound.) The n-th term has modulus $e^{-\pi n^2 t - 2\pi n \operatorname{Im} z} \le e^{-\pi n^2 t + 2\pi |n||z|}$. If $|n| \ge 4|z|/t$, then $2\pi|n||z| \le \pi n^2 t/2$, and the term is $\le e^{-\pi n^2 t/2}$. If $|n| < 4|z|/t$, the term is $\le e^{2\pi|n||z|} \le e^{8\pi|z|^2/t}$, and there are at most $8|z|/t + 1$ such n. Hence

$$|\Theta(z \mid \tau)| \le \Big(\frac{8|z|}{t} + 1\Big) e^{8\pi|z|^2/t} + \sum_n e^{-\pi n^2 t/2} \le A e^{B|z|^2}.$$

The series converges uniformly on compact sets by the same bound, so Θ is entire (Theorem 5.2) of order ≤ 2.

Step 2. (Quasi-periodicity.) Replacing n by $n+1$ in the sum,

$$\Theta(z+\tau \mid \tau) = \sum_n e^{i\pi(n^2+2n)\tau} e^{2\pi i n z} = e^{-i\pi\tau} e^{-2\pi i z} \sum_n e^{i\pi(n+1)^2\tau} e^{2\pi i (n+1) z} = e^{-i\pi\tau - 2\pi i z}\, \Theta(z \mid \tau).$$

Step 3. ($\Theta \not\equiv 0$.) On the real line the series converges uniformly, so $\int_0^1 \Theta(x \mid \tau)\, dx = 1$ (only $n = 0$ survives). Pick z_0 with $\Theta(z_0 \mid \tau) \neq 0$ and write $y_0 = \operatorname{Im} z_0$.

Step 4. (Lower bound.) Let $z_m = z_0 + m\tau$. Since $|e^{-i\pi\tau - 2\pi i z}| = e^{\pi t + 2\pi \operatorname{Im} z}$ and $\operatorname{Im} z_j = y_0 + jt$, Step 2 gives

$$|\Theta(z_m \mid \tau)| = |\Theta(z_0 \mid \tau)| \prod_{j=0}^{m-1} e^{\pi t + 2\pi(y_0 + jt)} = |\Theta(z_0 \mid \tau)|\, e^{\pi t m^2 + 2\pi y_0 m}.$$

If $|\Theta| \le A e^{B|z|^\rho}$ with $\rho < 2$, then, since $|z_m| \le |z_0| + m|\tau|$,

$$\pi t m^2 + 2\pi y_0 m + \log|\Theta(z_0 \mid \tau)| \le \log A + B(|z_0| + m|\tau|)^\rho,$$

which fails for large m. Hence the order is exactly 2. ∎

Exercise 5.11 (Picard's little theorem for finite order)

Show that an entire function of finite order that omits two distinct values is constant.

Solution.

Step 1. Let f have order $\le \rho$ and omit $a \neq b$. Then $g = f - a$ is entire and nowhere zero, and $|g| \le |f| + |a| \le (A + |a|)e^{B|z|^\rho}$, so g has finite order. By Hadamard's theorem (Theorem 5.1) with no zeros, $g = e^p$ with p a polynomial.

Step 2. Since f omits b, $e^{p(z)} \neq b - a$ for all z. As $b - a \neq 0$, there is $w \in \mathbb{C}$ with $e^w = b - a$.

Step 3. If p were non-constant, the fundamental theorem of algebra (Corollary 4.6) applied to $p - w$ would give z with $p(z) = w$, hence $e^{p(z)} = b - a$, a contradiction. So p is constant, and so is $f = a + e^p$. ∎

Exercise 5.13 (Solutions of $e^z = z$)

Show that the equation $e^z - z = 0$ has infinitely many solutions in $\mathbb{C}$.

Solution.

Step 1. $f(z) = e^z - z$ satisfies $|f| \le e^{|z|} + |z| \le 2e^{|z|}$, so its order is $\rho_0 \le 1$, and the integer k of Theorem 5.1 is 0 or 1. Also $f(0) = 1 \neq 0$.

Step 2. Suppose f had only finitely many zeros $a_1, \dots, a_N$. By Hadamard's theorem, $f = e^P \prod_{n \le N} E_k(z/a_n)$ with $\deg P \le k \le 1$. Each $E_k(z/a_n)$ is $(1 - z/a_n)$ times the exponential of a polynomial of degree $\le k$. Collecting factors,

$$e^z - z = Q(z) e^{\alpha z + \beta}$$

for a non-zero polynomial Q and constants $\alpha, \beta \in \mathbb{C}$.

Step 3. (Behaviour as $x \to -\infty$.) For real x, $|Q(x)e^{\alpha x + \beta}| = |Q(x)| e^{x \operatorname{Re}\alpha + \operatorname{Re}\beta}$, while $|e^x - x| = |x| + o(1)$ as $x \to -\infty$. If $\operatorname{Re}\alpha > 0$ the right side tends to 0; if $\operatorname{Re}\alpha < 0$ it grows faster than every power of $|x|$. Both contradict $|e^x - x| \sim |x|$. Hence $\operatorname{Re}\alpha = 0$.

Step 4. (Behaviour as $x \to +\infty$.) Then $|Q(x)e^{\alpha x + \beta}| = |Q(x)| e^{\operatorname{Re}\beta}$ grows at most polynomially. But $e^x - x \ge e^x/2$ for large x, a contradiction. So f has infinitely many zeros. ∎